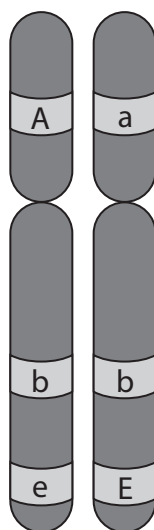


3 The nucleus of a cell contains chromosomes.

The diagram shows three genes present on a pair of chromosomes.



(a) (i) How many of the following statements about this diagram are correct?

- there are three gene loci
- the pair of chromosomes will be separated into different cells following mitosis
- one chromosome was inherited from the mother and one chromosome was inherited from the father

(1)

- ☐ **A** none
- ☐ **B** one
- ☐ **C** two
- ☐ **D** three

(ii) Describe what will happen to these chromosomes when they enter the interphase stage of the cell cycle.

(2)

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(b) Crossing over between pairs of chromosomes increases genetic variation.

(i) Name the stage of meiosis in which crossing over begins.

(1)

(ii) How many of the following combinations could result from one crossover of the chromosomes shown in the diagram?

- AbE
- ABe
- abE
- abe

(1)

☐ **A** one

☐ **B** two

☐ **C** three

☐ **D** four

(iii) Explain why alleles b and e are more likely to be inherited together than alleles A and e.

(2)

(Total for Question 3 = 7 marks)

